Paired Matched Terminal Supervision and Direction-Gain Allocation for Budgeted Sample-Column Interventions

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[Paper QR Code]

Tabular Prediction

Applications

Credit risk / healthcare / e-commerce / industrial analytics

Schema-semantic choices

Models interpret values using names + types.
Their schema embedding can be switched ON / OFF.
Each sample-column pair defines a candidate action.

Why local interventions?

The same schema embedding can help one sample and harm another. Nonlinear models can give it different effects in different feature contexts.

Challenge: learn each change's incremental value, then spend a limited intervention budget.

Two-stage Decision Framework

1. Static Default schema states per column; unacted samples keep Static.

2. Complement Flip one sample-column's schema state ON ↔ OFF; observed values and other column states stay unchanged.

3. Policy Select one winner per sample; rank equal-cost winners; execute top B%.

Stage 1 · Within-row Choose one candidate column → **Stage 2 · Across-row** Rank winners, execute top B%

	Col 1	Col 2	Col 3
Static:	·	·	×
Sample 1	×	·	×
Sample 2	·	×	×
Sample 3	·	·	·
Sample 4	×	·	×

Shaded = proposed row winner · ON × OFF

Execute top B% of row winners

Study

Outcome: Classifier's true-class NLL.
Lower NLL = Higher true-class probability.

7 UCI datasets
Online Shoppers · WDBC · Adult · Credit Default
Letter Recognition · Dry Bean · Steel Plates Faults

× **2 regimes**
BASE: no added corruption
COMPOUND: 30% mask + 10% training-label noise

× **5 folds**

× **5 fixed seeds**

Metrics Gain ↑, ΔF1 ↑, NDCG@1 ↑, Harm ↓ — Is Better

Paired Matched Terminal Supervision

Credit for each local change

Paired matched terminal supervision measures each candidate's incremental value against Static.
The Static–Complement true-class NLL difference supplies signed gain credit and a beneficial-direction target.

Matched Terminal Credit
NLL(Static) - NLL(Complement)

→ **PairedSign**
Target: Benefit (Yes/No)
Score: Likelihood of benefit

PairedMagnitude
Target: Continuous signed credit
Score: Expected incremental gain

Why matched credit?

Absolute outcomes can mix sample difficulty with intervention value. A row-matched Static reference targets incremental change.

One credit, two targets

Magnitude keeps gain size and sign.
Sign retains beneficial direction.
The same credit supports two learning priorities: continuous improvement vs. beneficial direction.

Learning and use

Downstream classifier → terminal NLL.
Paired credit → training targets.
Scorers (CatBoost / others) → candidate scores.
Predicted direction / gain → decisions.

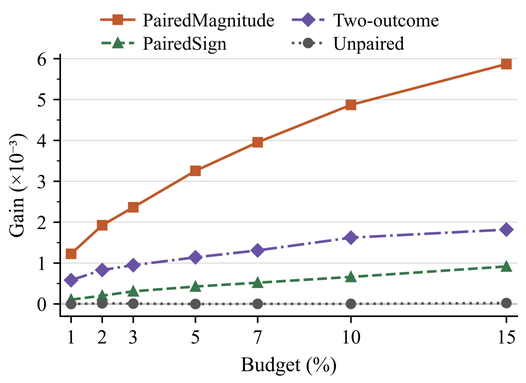
Supervision controls

Unpaired	-NLL(Complement) only No Static reference
Two-outcome	Predict both NLLs Subtract predictions
ColumnMean	Column-average credit One prior per column
TargetPermutation	Shuffle credit targets Break feature–credit link

Table I · Supervision · CatBoost Stage 1 Values: mean ± SD ; Best mean: bold red

Supervision	Gain@5% (×10⁻³)	Harm@5% (%)	ΔF1@5% (×10⁻³)	NDCG@1
PairedMagnitude	**3.256 ± 4.277**	42.98 ± 10.41	**+1.000 ± 3.598**	0.243 ± 0.061
PairedSign	0.426 ± 0.825	**21.98 ± 10.83**	+0.349 ± 0.868	**0.279 ± 0.088**
Unpaired	-0.002 ± 0.074	54.59 ± 7.39	+0.005 ± 0.024	0.202 ± 0.055
Two-outcome	1.141 ± 1.573	48.10 ± 4.59	+0.677 ± 0.645	0.202 ± 0.055
ColumnMean	0.018 ± 0.472	48.91 ± 6.02	+0.246 ± 0.402	0.189 ± 0.048
TargetPermutation	-0.733 ± 1.229	53.99 ± 4.59	-1.815 ± 3.615	0.195 ± 0.051

Fig. 1 · Gain · CatBoost Stage 1 Values: mean



Results & insight

PairedMagnitude : Highest mean Gain / ΔF1; 2.85× Two-outcome Gain @5% ; more Harm than PairedSign.
PairedSign : Best NDCG / lowest Harm → Direction favors local ranking and lower action risk.
Unpaired ≈ 0 Gain; ColumnMean weak → Unmatched outcomes / column priors provide little executed Gain.
TargetPermutation : Negative Gain / ΔF1 → Breaking feature–credit correspondence harms decisions.

Results & insight

PairedMagnitude leads mean Gain at all 7 budgets.
Two-outcome stays lower; Unpaired ≈ 0.
Direct learning of continuous matched credit yields greater executed Gain across budgets.

Sign-to-Magnitude Direction–Gain Allocation

Direction and gain across two decision stages

Direction prioritizes likely benefit; gain compares expected improvement. Stage 1 uses PairedSign to select one winner per sample.
Stage 2 uses a dedicated random-forest estimator to predict PairedMagnitude credit and rank winners across rows.

Stage 1 · Direction / PairedSign
Within-row candidate scoring. Retain one winner per sample.

→ **Stage 2 · Gain / PairedMagnitude**
Dedicated winner estimator. Across-row ranking and top-B% execution.

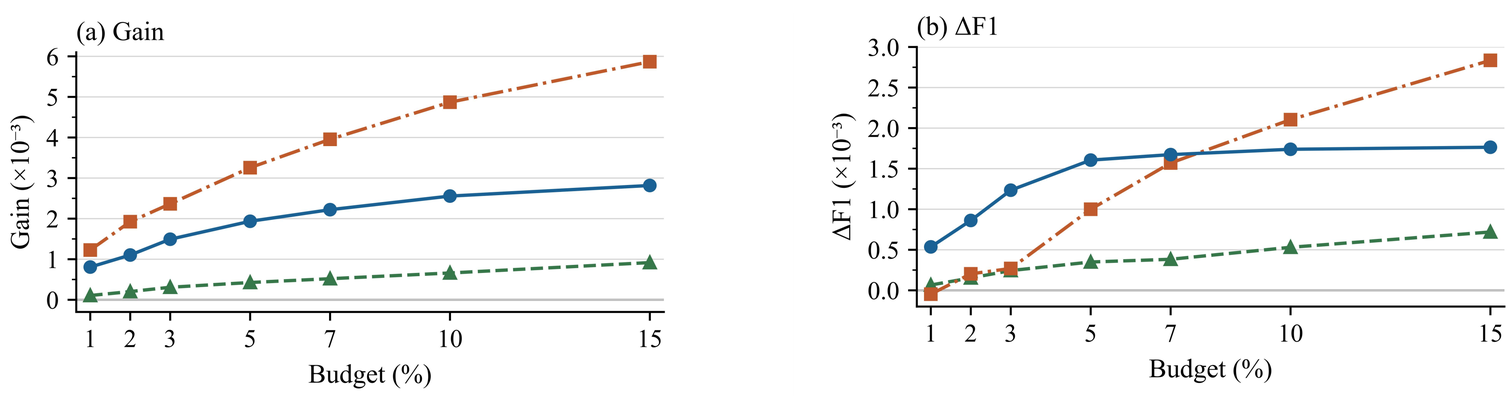
Table II · Policies · CatBoost Stage 1 Values: mean ± SD

Policy	Gain@5% (×10⁻³)	Harm@5% (%)	ΔF1@5% (×10⁻³)
Sign-only	0.426 ± 0.825	**21.98 ± 10.83**	+0.349 ± 0.868
Magnitude-only	**3.256 ± 4.277**	42.98 ± 10.41	+1.000 ± 3.598
Sign-to-Magnitude	1.935 ± 2.602	29.86 ± 7.25	**+1.606 ± 2.304**

Table III · Sign-to-Magnitude · Cross-Carrier

Carrier	Gain@5% (×10⁻³)	Harm@5% (%)	ΔF1@5% (×10⁻³)
HGB	1.550 ± 1.827	29.75 ± 6.63	+1.141 ± 1.569
CatBoost	**1.935 ± 2.602**	29.86 ± 7.25	+1.606 ± 2.304
PairLogit	1.831 ± 2.293	30.66 ± 6.89	**+1.676 ± 2.169**
LambdaMART	1.868 ± 1.910	**29.01 ± 7.35**	+1.476 ± 1.788

Fig. 2 · Gain and ΔF1 · CatBoost Stage 1 Values: mean



Results & insight

Sign-to-Magnitude (STM) · CatBoost at 5% :
STM vs. Magnitude-only : Harm ↓ / mean ΔF1 ↑ / Gain ↓ .
STM vs. Sign-only : Harm ↑ / mean ΔF1 ↑ / Gain ↑ .
Across four Stage-1 carriers, STM shows similar-scale 5% Gain, Harm, and ΔF1.

Results & insight

Mean Gain : Magnitude-only > Sign-to-Magnitude > Sign-only at all 7 budgets.
Mean ΔF1 : Sign-to-Magnitude leads at 1–7%; Magnitude-only at 10% / 15%.
The policies show different Gain–F1 balances; preference depends on budget.

Discussion

1. Probability gains may leave F1 unchanged. Gain tracks true-class probability shifts; F1 depends on predicted-class changes and confusion structure. A smaller NLL gain may therefore affect F1 more.

2. Practical value depends on cost and risk. Small average reductions may matter for score-based decisions; benefit must justify intervention costs and harmful actions. Harm (% of executed actions with worse NLL) motivates risk-aware allocation.

Limitations

1. Controlled actions simplify real systems. Single-column, equal-cost actions give an initial formulation. Real systems are more complex: interventions may interact, costs differ and operational constraints apply.

2. Matched feedback and deployment need validation. Offline paired outcomes can be costly or unavailable. Benchmark datasets and controlled corruption limit external validity; real-world validation is needed.

Conclusion

1. Matched credit yields complementary targets. Static–Complement NLL differences give incremental credit. PairedMagnitude retains signed gain size; PairedSign retains beneficial direction, with different execution and ranking trade-offs.

2. Direction guides choice; gain guides the budget. Sign-to-Magnitude combines these roles. With CatBoost at 5%, lower Harm and higher mean ΔF1 than Magnitude-only, despite lower Gain, support stage-specific use.

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